Zhiyin Yang

Ph.D. in Chemistry (UCLA) | Battery R&D | 2x Applied Materials Internships | 20 Publications
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CAREER SUMMARY

Ph.D. in Chemistry with 5 years' experience in energy storage materials, battery assembly, and thin-film deposition. Interned in battery and semiconductor divisions at Applied Materials.

EDUCATION

University of California, Los Angeles Sep. 2021 – March 2026

Ph.D. in Chemistry, GPA:3.98

- Advisor: Prof. Richard B. Kaner
- Awards: Division of Graduate Education Award, Chemists Association Excellence in Research Award

SKILLS

Battery R&D: Li-ion, Na-ion, Zn-ion, Electrode design, Coin/Pouch cell assembly, Pre-lithiation

Fabrication: Roll-to-roll, CVD, ALD, PVD, Laser Scribing, Glovebox, Spin Coating

Characterization: SEM, TEM, XRD, XPS, TGA, FTIR, Raman, UV-Vis, NMR, HPLC

Software & Data Analysis: OriginLab, MATLAB, Python, ImageJ

Battery Testing: Neware battery cyclers, Biologic electrochemical workstations, Electrochemical testing (CV, EIS, GCD)

Strengths: Design of experiment, scale-up, cross-functional collaboration, rapid prototyping

WORK EXPERIENCE

Applied Materials, Inc. Jun. 2025 – Sep. 2025

Chemist, Summer Intern, SPG (Semiconductor Product Group), Chemistry COE Team *Santa Clara, CA*

- Developed selective Cu capping films via CVD/ALD using Group 5–7 metal precursors (V, Nb, Cr, Mo, W, Mn).
- Achieved high selectivity on Cu/dielectric interfaces, validated by XPS and TEM.
- Collaborated with materials, metrology, and integration teams to ensure manufacturing.

Applied Materials, Inc. Jun. 2023 – Sep. 2023

Process Engineer, Summer Intern, CTO office, Li-ion Battery Team *Santa Clara, CA*

- Developed roll-to-roll pre-lithiation method for Si/graphite anodes.
- Optimized printing parameters for industrial scale-up, improving Initial Coulombic Efficiency by ~15%.
- Coordinated with automation and testing teams on pilot line implementation.

RESEARCH EXPERIENCE

Silicon/graphene Composite for Fast-Chargeable Li-ion Batteries Jan. 2024 – Present

- Designed a laser-scribing route to synthesize Si/graphene composites at room temperature.
- Developed rapid reprecipitation method to integrate Si microparticles into a 3D conductive network.
- Enabled 3× faster processing and improved structural and mechanical integrity of the composite electrode.
- Delivered high capacity (>1000 mAh/g) and stable cycling over 200 cycles.

V_2O_5/Polymer Hybrids and Layered Electrodes for Na-ion Batteries Jan. 2023 – Feb. 2025

- Engineered molecularly confined V_2O_5/aniline trimer hybrid to enhance electronic conductivity.
- Synthesized Fe/Mg-doped layered Na-ion cathodes with improved structural stability.
- Achieved >90% capacity retention over 1000 cycles at 2C rate.
- Developed a two-step laser heating strategy reducing energy and time cost by over 50%.

Aqueous Electrolyte for Zn and Li-ion Batteries Nov. 2021 – Feb. 2023

- Formulated ultra-concentrated aqueous LiTFSI electrolyte suppressing water activity and gas evolution.
- Improved Zn-ion cathode kinetics using laser-scribed V_2O_5 composite scaffold.
- Demonstrated safe, non-flammable, high-rate operation with >92% capacity retention at 12 C rate.
- Applicable for cost-effective, scalable, and safe grid-level energy storage systems.

PUBLICATIONS

- 20 Publications, 5 first author, 3 second author, total citation: 220 2020 – Present
- Served as a reviewer for 10+ scientific journals (*J. Am. Chem. Soc, ACS Appl. Mater. Interfaces*, etc.) 2023 – Present